Question 7. You have been employed at Kapasa Makasa farm to manage the horticultural section. However, you discover that the only fertilizer available in stores is basal dressing, what horticultural crops could you grow to make use of the fertilizer and at least without making the farm make a loss.

(3 marks)

Question8. Vegetables are classified according to the audible parts. Give examples of vegetables that can be classified in the category of:

a) Flowers

b) Seeds

c) Stems

d) Roots

(8 marks)

Question 9. Explain the role of potassium in the growth of cabbage crop. **(2 marks)**

Question 10. The field visit that you did as a practical reviewed that rape was infested with red spider mites. Can you explain the control method that is recommended to control the insect pestsconsidering safetyof the people that live surrounding the field. **(3 marks)**

Question 11. Why should integrated pest management (IPM) be practiced in modern agriculture?

(3 marks)

Question12.There has been an increase in the quantity of flowers imported from Zambia by European countries. What could contribute to the increase in demand of cut flowers?

(2 mark)

Question13. Explain the benefits of producing cut flowers in the green house compared to open fields.

(2 marks)

Question14. The practical we did to propagate the oranges using lemon rootstalk, it was discovered that after **2(two)** months the scion did not shoot. According to your observation during the practical what do you think contributed to the failure in terms of material used and other factors?

(4 marks)

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Department of Agriculture and Aquatic Sciences

SESSIONAL EXAMINATIONS 2022

COURSE CODE: NR 310

COURSE NAME: CROP SCIENCE AND HORTICULTURE

TIME ALLOWED: THREE (3) HOURS

DATE: 17TH OCTOBER 2022

INSTRUCTIONS: 1. Answer the ALL the questions

2. The exam has a total of 60 marks

Course Code: NR 310

Course name: Crops Science and Horticulture

Duration: 3 hours

Question1 What are some benefits of producing vegetables in the green house compared to open field production. **(3 marks)**

Question 2 What determines the mobility of the nutrient in the plant? Give examples of mobile and non-mobile nutrients. **(4 marks)**

Question 3. Mr Musenzo measured that each orange plant in the field occupied the space of $4.5m^2$. using this information, answer the following questions.

a) The spacing between the plants (intra-row spacing) if the spacing between the beds (inter- row spacing) is 3metres. **(4 marks)**

b) Estimate the number of orange plants in Mr. Musenzo's field measuring 35m X 25m. **(3 marks)**

c) Mr. Musenzo buys **50kg** bag of basal fertilizer, how much can be applied per plant? **(4 marks)**

Question4.

a) Explain what you can do to successfully carry out orange propagation using lemon rootstalk. **(3 marks)**

b) The farmer is producing the rose flowers in the soil give reasons why as an expert you would encourage that practice? **(2 marks)**

Question 5. What role do the following factors play in the growth of vegetables?

a) Temperature

b) Humidity

c) Light

d) Moisture **(8 marks)**

Question 6. In the field the farmer wants to know the size of the maize leaf. How can you assist the farmer to get accurate measurement of the leaf. **(2 marks)**

KAPASA MAKASA UNIVERSITY

Department of Agricultural and Aquatic Sciences

Course Code: NR 310

Course name: Crops science and Horticulture

Duration: 3 hours

1 Why is the soil beneficial in the production of vegetables? **6 Marks**

2 It was discovered that the plant has got **only** a tap root to a depth of 2 metres without any fibrous roots what bad effect could that posse to the plant and what would be the possible benefit to the plant? **4 Marks**

3 Mention soil conditions that can reduce root growth in vegetable production **2 Marks**

4 The tractor ploughing in the field had caused compaction to the soil why should be compaction controlled in the flower production? **6 Marks**

5 What are some factors that determine the survival of crops in Muchinga province? **4 Marks**

6 How would plants react to unfavourable conditions such as:

(a) Water stress

(b) Excess water

(c) Wind

(d) Shade

(e) High temperature **10 Marks**

7.Explain the role of Nitrogen in the growth of cabbage crop **4 Marks**

8 Which propagation methods can you recommend to a farmer who wants to establish a horticultural production farm for vegetables and flowers? **6 Marks**

9 The field located in the middle of housing complex surrounded by workers houses is facing insect infestation Can you explain the control method that is recommended to control the insect pests **4 Marks**

10 Why is integrated pest management (IPM) important in modern agriculture? **6 Marks**

The elder sister wants to open up the farm to produce rose flowers in Chinsali for the Lusaka market, what advice can you provide to her? **8 Marks**

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KAPASA MAKASA UNIVERSITY CAMPUS

DEPARTMENT OF AGRICULTURE AND AQUATIC SCIENCES

SUPPLEMENTARY EXAMINATION FOR 2018/19 ACADEMIC YEAR

COURSE NAME: **HORTICULTURE** CODE: **NR 310**

DATE : NOVEMBER 9, 2019

TIME : THREE (3) HOURS TOTAL MARKS: 100

Instructions:

1. There are three sections in this examination.
2. Section A has two compulsory questions and each question is 20 marks.
3. Answer two (2) questions each from Sections B and C. Each question is 15 marks.
4. There are 100 marks allocated to this paper
5. Time allowed is three (3) hours.
6. Please write legibly and include all the necessary steps in your calculations in order to get full marks.
7. Maintain **four decimal places** during calculations till the final answer. After finding the final answer, record it **without a decimal place**.
8. Please be concise (short, clear and to the point) in your explanations; marks will not be given for waffling!
9. When answering questions, follow further instructions in the Question paper and Answer booklet.

SECTION A

1. The section contains two compulsory questions which carries 20 marks each.
2. Show all your workings for sub-questions that require calculations as no mark(s) will be given for any unclear working/calculations even if the final answer may be correct.

QUESTION 1

a. Protected agriculture is defined as the modification of the natural environment to suit crop growth and extend the growing season. Outline any **three** challenges that have led to low adoption of protected agriculture technologies in Zambia. **[6 marks]**

b. Supposing ZARI 2020 a dwarf high yielding maize variety is suitable in any protected agriculture technique. The variety records 100% plant emergence, produces **two** cobs per plant and yields 15 tons in any technique which measures 100 m by 50 m. The yield is only possible when the variety is planted at intra-row spacing of 45 cm and inter-row spacing of 90 cm.

(i) Calculate the number of planting rows basing on 100 m length. **[2 marks]**

(ii) Calculate the number of planting stations each planting row will have. **[2 marks]**

(iii) Calculate the number of cobs that will be harvested in a technique which measures 100 m by 50 m. **[4 marks]**

(iv) List any **six consequences** to the crop if the recommended plant population to obtain 15 tons is 25, 000 plants per hectare. **[6 marks]**

QUESTION 2

One major challenge of producing exotic vegetables in Zambia is gluts during marketing.

(a) Define gluts. [2 marks]

(b) Explain in details **any two** causes of gluts. [5 marks]

(c) Supposed you were approached by vegetables growers in your district, suggest **three** ways on how you would solve the challenge of gluts. **[6 marks]**

The success of most vegetable production depends on nursery management.

(d) Define a nursery [2 marks]

(e) Suppose a farmer with intentions to grow **green beans** approaches you and tells you he wants to put up a nursey first and transplant latter in the fourth week. How will you guide the farmer? [5 marks]

SECTION B : CROP SCIENCE (ANSWER ANY TWO QUESTIONS)

Each question carries 15 marks.

QUESTION 1

Susan Nsofwa is an agronomist for ZAMSEED stationed in Shiwangandu District. She planted a trial for a new maize variety with maturity period of 130 days. Her maize field received solar energy of about 26,953 million KJ ha^{-1} and harvested her maize at moisture content of 17%. The maize variety had an ash content of 10%, total plant dry matter of 18, 170 $kgha^{-1}$ (comprising of DM_{Root} = 3,950 $kgha^{-1}$ and DM_{Stover} = 7,280 $kgha^{-1}$) and estimated that the plant used 25% of the photosynthates for maintenance. Given that 15,792 KJ of energy is required to produce a kg of glucose.

a. How many tons of maize did she produce per hectare at moisture content of 12.5%? [3 marks]

b. Supposing she wanted to sell her maize to FRA at 12.5% moisture content, how many bags will she sell? [2 marks]

c. Calculate,

(i) $DM_{Economic\ yield}$ [2 marks]

(ii) DM_{Actual} [2 marks]

(iii) DM_{Net} [2 marks]

d. How much solar energy was converted to chemical energy by the maize per hectare? [2 marks]

e. Calculate the photosynthetic efficiency of the maize variety Susan Nsofwa planted. [2 marks]

QUESTION 2

a. Several factors affect photosynthesis in plants. Outline **any three** external factors. **[6 marks]**

b. Write the centres of origin and scientific names for the following crops;

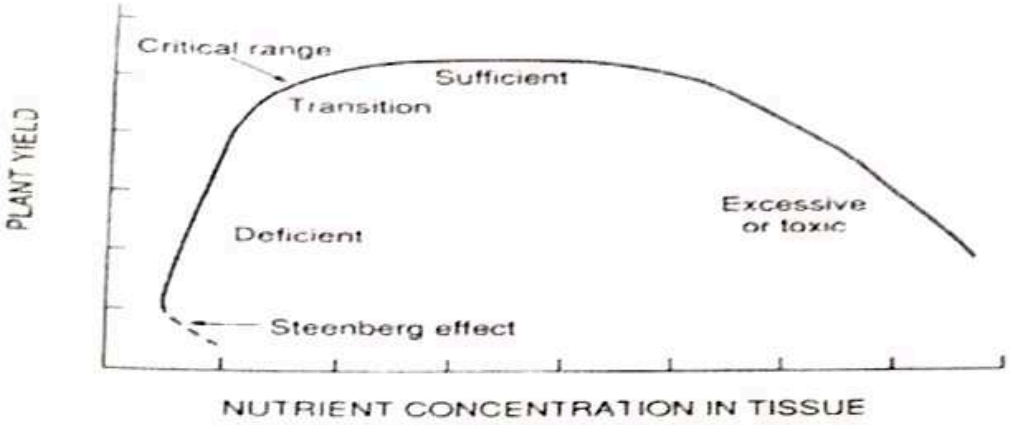
(i) Cassava **[3 marks]**

(ii) Sunflower **[3 marks]**

(iii) Maize **[3 marks]**

QUESTIONS 3

a. Study the graph below carefully before answering the questions.



(i) Explain the relationship between nutrient concentrations in the plant tissue and pant yield or growth as illustrated by the graph above when nutrient concentration reaches Steenberg effect. [2 marks];

(ii) Which of the nutrient concentration in plant tissue encourages luxury consumption and why? **[4 marks]**

b. Gibberellins is a phytohormone which affects plant growth and development.

(i) Explain how it is transported after it has been synthesised. **[2 marks]**

(ii) What plants organs synthesis the hormone? **[2 marks]**

c. Crop yields are determined by yield components and all plants will always compensate any reduction in formation of these yield components imposed by photosynthates limitation in the subsequent yield component formation in the next growth stage. List **any five** yield components of fruit crops (tomatoes). **[5 marks]**

SECTION C: HORTICULTURE (ANSWER ANY TWO QUESTIONS)

1. Each question carries 15 marks
2. Show all your workings for questions that require calculations as no mark(s) will be given for any unclear working/calculations even if the final answer may be correct

QUESTION 1

(a) Land preparation is an important role in realising the full potential of vegetable crops. Supposing a vegetable grower approaches you for an advice on how to prepare his land, what advice will you give if he tells you transplanting will be done in;

(i) The rainy season **[2 marks]**

(ii) A garden with high water table which is close to soil surface. **[2 marks]**

(iii) A cool dry season **[2 marks]**

(iv) A garden with low water table which is close to soil surface. **[2 marks]**

(b) Companion planting is highly advocated by most horticulturalists in olericulture

(i) What is meant by companion planting? **[3 marks]**

(ii) Give **two** reasons for using companion planting. **[2 marks]**

(iii) Name **two** crops that can be companions with cucumbers. **[2 marks]**

QUESTION 2

Chilala carried a vegetable market survey and found that onions are on high demand in Chinsali throughout the year. Based on the survey results, he goes in onion production and partitions his garden (**1 ha**) in two equal portions (Each having a length of 100 m and 50 m width). He used 10 cm and 25 cm as intra- and inter – row spacing, respectively on the Northern portion; and 20 cm and 35 cm as intra- and inter – row spacing, respectively for the Southern portion. The onion planting was based on market segmentation.

a. What planting method should be employed by Chilala if needs to sale the onion quickly? Give a reason for your answer. **[4 Marks]**

b. What type of customers did he target in the Northern portion? Give a reason for your answer. **[6 Marks]**

c. Supposing Chilala opted to use Compound D (10:20:10) and urea (46%N) as basal and top dressing respectively. Comment on his quality of his onion. **[5 Marks]**

QUESTION 3

a. During a horticulture conference, a DACO of Horizon district in the New Found Land Country said, "Using tunnels, fruit production cannot be improved in any country!"

(i) Do you agree to the DACO's statement? Justify your answer. **[3 marks]**

(ii) Supposing you were in attendance in the conference, suggest **three ways** to encourage the use of tunnels in the New Found Land Country. **[6 marks]**

b. Rosa indica L is a highly valuable crop in Zambia. The yield of the crop is severely affected by insects and diseases. Briefly explain how one would identify and control the following;

(i) Aphids. **[3 marks]**

(ii) Rose Rosette Disease. **[3 marks]**

END OF EXAMINATION